KK–L–156
MAY 16, 1956
SUPERSEDING
Int. Fed. Spec. KK–L–00171d (COM–NBS)
16 December 1954, and
Fed. Spec. KK–L–171c
31 July 1950

FEDERAL SPECIFICATION

LEATHER, CATTLEHIDE, HARNESS, VEGETABLE-TANNED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers harness leather made from cattlehides and primarily used for harness straps.

1.2 Classification.

1.2.1 *Selections.*—The harness leather covered by this specification shall be furnished in the following selections, as specified:

Selection A.
Selection B.
Selection C.

1.2.2 *Types.*—The harness leather shall be furnished in the following types, as specified:

Type I.—Treated for resistance to mildew.
Type II.—Untreated.

1.2.3 *Form (trim).*—The harness leather shall be furnished in the form of sides or backs, as specified. (See 3.8.)

1.2.4 *Weights.*—The harness leather shall be furnished in the following weights, as specified:

Extra light
Light
Medium
Heavy
Extra heavy

1.2.5 *Color.*—The color of the harness leather shall be black or russet, as specified.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

KK–L–311—Leather; Methods of Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Specifications and standards applicable only to individual departments are listed in section 7.

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3. REQUIREMENTS

3.1 Material.—The leather shall be made from cattlehides, except that no bull hides shall be used.

3.2 Tannage.—The hides shall be tanned by a process commercially known as vegetable tanning.

3.3 Finish.—The leather shall be full grain or full grain lightly snuffed. The flesh side shall be smooth and free from flesh and may be shaved or skived to secure greater uniformity of thickness.

3.4 Cracking.—The leather shall not crack when tested as described in 4.3.3.1.

3.5 Tensile strength and elongation.—The tensile strength shall be not less than 4,000 pounds per square inch. The elongation shall be not more than 20 percent at a load of 2,500 pounds per square inch. (See 4.3.3.2.)

3.6 Chemical requirements.—The leather shall conform to the chemical requirements in table I.

TABLE I.—*Chemical requirements (moisture-free basis)*

Characteristics	Minimum	Maximum
Chloroform extract (Grease):		
Black leather .....percent	20.0	35.0
Russet leather .... do	15.0	30.0
Water soluble ..... do	...	18.0
Total ash ..... do	...	5.0
Degree of tannage	40.0	75.0
Acidity (pH)	3.3	4.5
Paranitrophenol (type A only)	0.20	0.45

3.7 Selections.

3.7.1 *Selection A.*—Each unit of leather shall be free from deep cuts, damaged grain, open scratches, grubs, brands, hip marks, and rough shoulders, or any other defects which affect cutting value.

3.7.2 *Selection B.*—Each unit of leather may contain imperfections listed in selection A, but such imperfections shall not show in cutting value a greater loss than 6 percent of the total area of the unit.

3.7.3 *Selection C.*—Each unit of leather may contain imperfections listed in selection A, but such imperfections shall not show in cutting value a greater loss than 12 percent of the total area of the unit.

3.8 Trim.—Sides and backs shall comply with the definitions for sides and backs in Federal Specification KK–L–311.

3.9 Thickness.—The average thickness of each unit of leather in the sample for test shall be within the limits specified in table II for the respective weights.

TABLE II.—*Thickness*

Weight	Average thickness
	64ths inch
Extra light	8 and below 10
Light	10 and below 12
Medium	12 and below 14
Heavy	14 and below 16
Extra heavy	16 and above

3.10 Mildew resistance.—When type I leather is specified, the leather shall be treated with paranitrophenol so as to contain not less than 0.20 percent nor more than 0.45 percent, calculated on a moisture-free basis.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Formation of inspection lots.*—Lots of each selection, type, form, weight, and color shall be formed as described in 3.1 of section 6 of Federal Specification KK–L–311.

4.1.2 *Sampling for inspection.*—Samples shall be taken for inspection as described in 3.2 of section 6 of Federal Specification KK–L–311.

4.1.3 *Sampling for physical and chemical tests.*—Samples for physical and chemical tests shall be taken as described in 3.3.1 and 3.3.2 of section 6 of Federal Specification KK–L–311.

4.2 Inspection.

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4.2.1 Each unit in the sample shall be inspected for finish, selection, color, trim, and thickness.

4.2.2 *Thickness.* — The thickness shall be determined as described in 4.1.1 of method 1011 of Federal Specification KK–L–311 for hides, skins, kips, and similar materials.

4.3 Tests.

4.3.1 *Specimens for physical tests.*—Specimens for physical tests shall be taken from the test area shown in figure 6, section 6, of Federal Specification KK–L–311 as required in the applicable method of test.

4.3.2 *Specimens for chemical tests.*—Specimens for chemical tests shall be taken from the composite sample prepared as described in method 6002 of Federal Specification KK–L–311, as required in the applicable method of test.

4.3.3 *Methods of test.*

4.3.3.1 *Cracking.* — The resistance of the leather to cracking shall be determined as described in method 4011 of Federal Specification KK–L–311. The sizes of the mandrels used for testing different weights shall be as follows: Extra light, 5/16 inch; light, 6/16 inch; medium, 7/16 inch; heavy, 8/16 inch; and extra heavy, 9/16 inch.

4.3.3.2 *Other tests.* — Other physical and chemical tests shall be conducted as described in the following methods of Federal Specification KK–L–311:

Test	Method No.
Tensile strength	2011
Elongation	2021
Moisture	6221
Chloroform extract	6311
Water-soluble material	6341
Insoluble ash	6411
Total ash	6421
Nitrogen	6441
Acidity (pH)	6621
Combined tannin	6631
Degree of tannage	6641
Paranitrophenol (type I only)	6711

4.4 Acceptance of lot.—A lot of leather shall be considered as meeting the requirements of this specification if the conditions described in 5.1 of section 6 of Federal Specification KK–L–311 are fulfilled.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.—Unless otherwise specified, commercial packaging and packing for domestic shipment — immediate use, domestic shipment and storage, or overseas shipment, will be acceptable (see 6.1).

5.2 Marking.—Unless otherwise specified, shipping containers shall be marked with the selection, type, form, weight, and color of the material as defined by the contract or order under which the shipment is made, the quantity of each type and size contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following, and should exercise any desired options offered herein.

(a) Title, number, and date of this specification.

(b) Selection, type, form, weight, and color required (1.2).

(c) Whether the subject commodity is to be packaged and packed for domestic shipment (immediate use), domestic shipment (storage), or overseas shipment, 5.1, as follows:

(1) *Domestic shipment (immediate use).* —Applies to that material required for immediate use without storage or reshipment by the initial domestic receiving activity.

(2) *Domestic shipment (storage).*—Applies to that material for storage and reshipment to a domestic receiving activity not involving sea transportation.

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(3) *Overseas shipment.*—Applies to that material to be transported by ocean-going or coastwise vessel and handled, issued, and stored in the military supply system.

(d) Marking (see 5.2).

6.2 Transportation description.—The transportation description applicable to these items is:

Leather, not otherwise indexed by name.

Carload weight 24,000 pounds subject to Rule 34, Uniform Freight Classification.

Truckload weight 24,000 pounds.

6.3 A bull hide usually ranges in weight from 75 to more than 100 pounds in the green, salted state. It is characterized by very thick and rough neck and shoulders. It is excluded from the class of cattlehide to be used for harness since it does not produce good quality leather.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard:*

Federal Specifications:

V–T–276—Thread; Cotton.

NN–B–621—Boxes, Wood, Nailed and Lock-Corner.

QQ–S–781—Strapping, Flat; Steel.

UU–P–268—Paper, Kraft, Wrapping.

CCC–B–811—Burlap, Jute.

CCC–O–721—Osnaburg; Cotton.

DDD–S–751—Stitches; Seams; and Stitching.

PPP–B–585—Boxes, Wood, Wirebound.

PPP–B–601—Boxes; Wood, Cleated-Plywood.

Military Specifications:

JAN–P–103—Packaging and Packing for Overseas Shipment — Boxes: Wood Cleated: Solid Fiberboard.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes: Wood Nailed.

JAN–P–125—Packaging and Packing for Overseas Shipment — Barrier Materials Waterproof, Flexible.

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.3 Application.—Leather shall be packaged and packed for domestic shipment (immediate use), domestic shipment (storage), or overseas shipment, as specified.

7.4 Domestic shipment (immediate use).—Leather shall be packed in containers which (together with packaging, wrapping, and inner packing) are acceptable by common or other carriers for safe transportation to point of destination specified in shipping instructions at the lowest transportation rate for such supplies.

7.5 For domestic shipment (storage).

7.5.1 *Packaging.*—Twelve, 24, or 36 hides, as specified in the invitation for bids or contract, shall be packaged in bundles for domestic shipment as follows:

7.5.1.1 Each set of six hides shall be reversed so that the heads and butts, backbones, and bellies are adjacent to the next set, and

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then neatly and tightly rolled together and wrapped in kraft wrapping paper of not less than 80 pounds basis weight (24 by 36—500) conforming to grade B, No. 2, of Federal Specification UU–P–268. The paper shall be folded to enclose each end of the roll completely and shall be secured with gummed kraft tape.

7.5.1.2 In addition, each roll of hides shall be protected by overwrapping in one of the following materials:

(a) Single-faced corrugated fiberboard in which the corrugated paper is not less than 0.009 point sulphate kraft and the linerboard is of not less than 47 pounds bursting strength (Mullen or Cady).

(b) Solid fiberboard of not less than 0.030 point caliper. The wrap shall extend not less than one inch beyond the ends of the roll, and shall completely encircle the roll, overlapping 4 inches. The wrap shall be secured in place by tying with suitable cord twice around the body approximately 12 inches from each end, and once lengthwise, tying with nonslip knots.

7.5.2 *Packing.*—Rolls of hides, packaged as described in 7.5.1, shall be inserted in cloth covering material which has been formed into bias-sewn tubing and the tubing shall be pulled tightly around all sides. Material shall be cotton osnaburg, not less than seven ounces to the square yard, conforming to Federal Specification CCC–O–721, or jute burlap, 40-inch width, 10 ounces per linear yard, conforming to Federal Specification CCC–B–811. Bias-sewn tubing shall be made by placing the selvage edge of the cloth covering material to the opposite edge and stitching not less than ⅜ inch nor more than ¾ inch from the edge, using stitch type 401 and seam type SSa–1 or LSa–1 conforming to Federal Specification DDD–S–751. Not less than three stitches per inch, and not finer than 12/4 thread in needle and not finer than 16/3 thread in looper shall be used. Proper width of material shall be used so that the finished width of tubing shall snugly fit the rolls of hides. The degree of bias shall be not less than 50 degrees, and the angles used within this range shall remain reasonably constant throughout the sewing operation. Thread shall be type IA1 conforming to Federal Specification V–T–276.

7.5.2.1 Sealing of the tubular bales shall be by one of the following methods:

(a) Bales may be sealed at each end with two double-loop, 6-inch, 16-gage galvanized wire ties. The first wire tie shall be applied as close to the bale as possible so that there will be a minimum of slack in the tube, the cloth being gathered evenly to form an ear, and all edges of the cloth being three to five inches from the wire tie at the base of the ear. The second wire tie shall be applied at a distance approximately one inch from the first wire tie. The twisted ends of the wire ties shall alternate and face in opposite directions.

(b) Bales may be sealed with one end presewn flat, seam to be one inch from the end of the precut tube, the other end to be closed with two wire ties.

(c) Hand-sewn bales will be permitted if specified by the procuring agency.

7.5.2.2 *Boxes.*—When specified in the contract, purchase order, or invitation for bids, hides packaged as specified in 7.4.1 shall be packed in an exterior shipping container constructed in accordance with Federal Specification PPP–B–601 (domestic type), NN–B–621, or PPP–B–585.

7.6 For overseas shipment.

7.6.1 *Packaging.*—Packaging shall be as specified in 7.4.1.

7.6.2 *Packing.*—Leather shall be packed in either bales or boxes as follows:

7.6.2.1 *Bales.*—Rolls of leather shall be packed in bales as specified in 7.4.2, except that in addition each roll shall be completely encased in an overwrap of waterproof kraft wrapping paper conforming to type B–2, of Military Specification JAN–P–125, and tightly folded to enclose each end, prior to inserting the roll into the cloth covering material.

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7.6.2.1.1 *Strapping of bales.*—Flat, nailless, painted steel straps, ⅝ inch wide by 0.018 inch thick, conforming to class A of Federal Specification QQ–S–781, shall be used for binding the bales. Straps shall be applied around the bale spaced not less than 10 inches from each end, avoiding knots or cord used for inner wrapping. Sufficient tension shall be applied to the flat metal strapping to insure tight application.

7.6.2.1.2 *Boxes.*—Alternatively, the rolls of leather wrapped in paper as specified, and then overwrapped in waterproof paper as specified in 7.6.2.1, may be packed in boxes constructed and strapped in conformance with the requirements of Military Specification JAN–P–103, Federal Specification PPP–B–601 (overseas type), or Military Specification JAN–P–106. Gross weight of each shipping box shall not exceed 100 pounds.

7.7 Marking of shipments.—Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129. The nomenclature shall be as follows: Leather, Cattlehides, Harness, Vegetable-Tanned.

MILITARY INTERESTS

Army—M O Q
Navy—Sh S MC
Air Force.

Copies of this specification may be purchased for ents each.

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